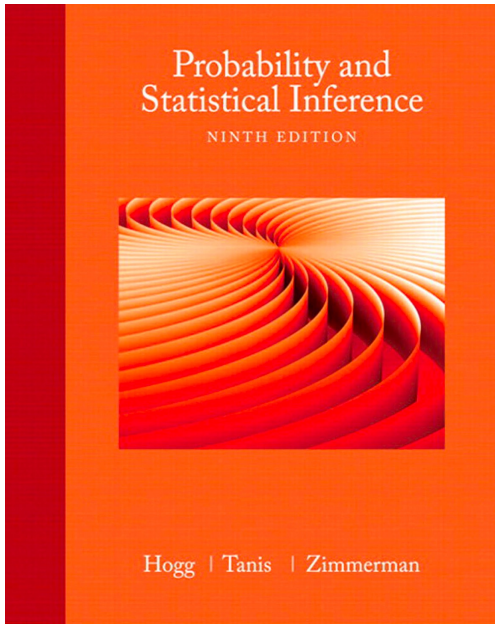
STA 2002, Summer 2023
Probability and Statistics II

Probability Review and Sampling Distribution

Probability Review

Chapter 1-5 in the book



- Some discrete distributions
- Some continuous distributions
- Expectation, variance, MGF
- Central limit theorem
- Law of Large Numbers

Basic terminology

- Random variable X follows distribution $f(x)$

$$X \sim f(x)$$

- ***i.i.d.*** *: Independent and identically distributed*

$$X_i \sim f(x), \qquad i = 1,2, \dots, n$$

- **Draw a sample X from a distribution $f(x)$ means $X \sim f(x)$.**

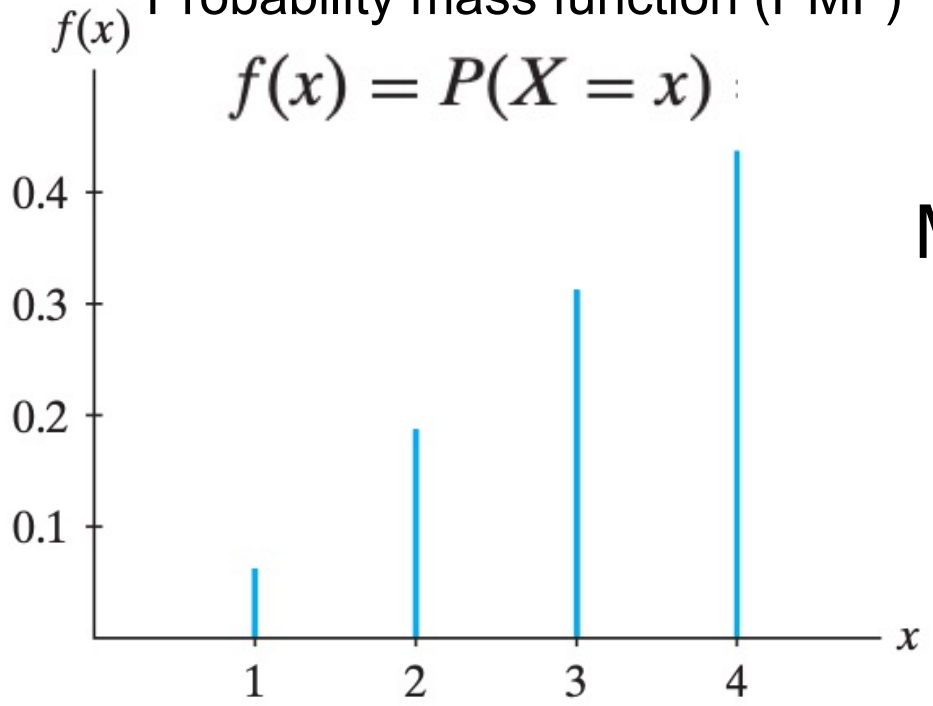
Discrete Distributions

Discrete Distributions

- When the outcome can only take discrete values (possibly infinite many)

Probability mass function (PMF)

$$f(x) = P(X = x)$$



CDF: $F(x) = \sum_{x_i \le x} f(x_i)$

Mean (expectation): $E(X) = \Sigma_i x_i f(x_i) = \mu$

Variance: $V(X) = \Sigma_i f(x_i)(x_i - \mu)^2 = \sigma^2$

$$V(X) = E(X^2) - [E(X)]^2$$

Bernoulli Distribution

$X \sim Bernoulli(p) \quad x = 0,1$ p: probability of success

Application: To model the outcome of experiment with two possible outcomes (Success = 1, Failure = 0).

e.g. Flip a coin:  **or**  **?**

pmf, mean and variance:

$$f(x) = p^x(1-p)^{1-x}, \qquad x = 0,1$$

$$\mu = E(X) = \sum_{x=0}^{1} x\,p^x(1-p)^{1-x} = (0)(1-p) + (1)(p) = p$$

$$\sigma^2 = \mathrm{Var}(X) = \sum_{x=0}^{1} (x-p)^2 p^x (1-p)^{1-x}$$

$$= (0-p)^2(1-p) + (1-p)^2 p = p(1-p)$$

Binomial Distribution

$X \sim Bin(n, p)$ $x = 0,1,\ldots,n$ p: probability of success

Application: To model the number of successes in ***n*** independent Bernoulli trials. *e.g. Flip n coins.*

pmf, mean and variance:

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, 2, \ldots, n$$

$$\mu = E(X) = np \qquad \sigma^2 = E(X^2) - [E(X)]^2 = np(1-p)$$

Fact: sum of *n* independent Bernoulli variables (*p*) is Binomial (*n*,*p*)

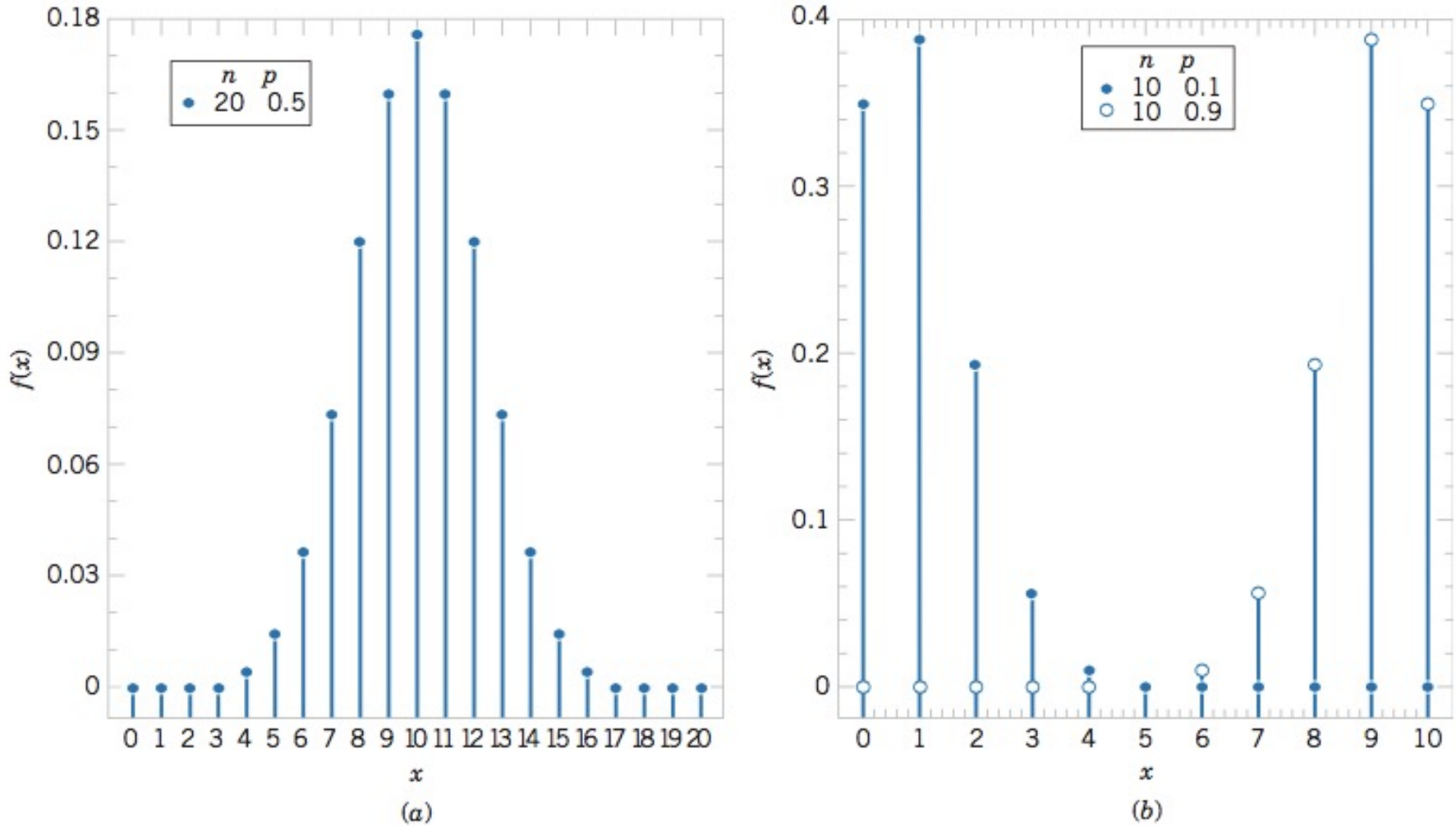


Figure 3-8 Binomial distributions for selected values of n and p.

Consider the following random experiments and random variables:

1. Flip a coin 10 times. Let X = number of heads obtained.
2. A worn machine tool produces 1% defective parts. Let X = number of defective parts in the next 25 parts produced.
3. Each sample of air has a 10% chance of containing a particular rare molecule. Let X = the number of air samples that contain the rare molecule in the next 18 samples analyzed.
4. Of all bits transmitted through a digital transmission channel, 10% are received in error. Let X = the number of bits in error in the next five bits transmitted.
5. A multiple-choice test contains 10 questions, each with four choices, and you guess at each question. Let X = the number of questions answered correctly.
6. In the next 20 births at a hospital, let X = the number of female births.
7. Of all patients suffering a particular illness, 35% experience improvement from a particular medication. In the next 100 patients administered the medication, let X = the number of patients who experience improvement.

Negative Binomial Distribution

$X \sim NB(r,p)$ $x = r, r+1, r+2, \ldots$ p: probability of success

Application: observe a sequence of independent Bernoulli trials *until exactly r successes occur*

pmf, mean and variance:

$$f(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}$$

$$\mu = E(X) = \frac{r}{p} \qquad \sigma^2 = \frac{rq}{p^2} = \frac{r(1-p)}{p^2}$$

Maclaurin's series expansion

$$\frac{1}{1-z} = \sum_{n=0}^{+\infty} z^n$$

$$\frac{1}{(1-z)^{r+1}} = \sum_{n=r}^{+\infty} \binom{n}{r} z^{n-r}$$

Geometric Distribution

$X \sim Geo(p)$ $\quad x = 1,2,3, \ldots$ $\quad$ p: probability of success

Application: observe a sequence of independent Bernoulli trials *until exactly r =1 successes occur, Negative binomial (1, p).*

pmf, mean and variance:

$$f(x) = p(1-p)^{x-1}, \qquad x = 1,2,3,\ldots.$$

$$\mu = E(X) = \frac{1}{p} \qquad \sigma^2 = \frac{q}{p^2} = \frac{1-p}{p^2}$$

Maclaurin's series expansion

$$\frac{1}{1-z} = \sum_{n=0}^{+\infty} z^n$$

$$\frac{1}{(1-z)^{r+1}} = \sum_{n=r}^{+\infty} \binom{n}{r} z^{n-r}$$

Hypergeometric Distribution

$$X \sim Hypergeometric(N, N_1, n) \quad 0 \le x \le n, x \le N_1, n - x \le N_2$$

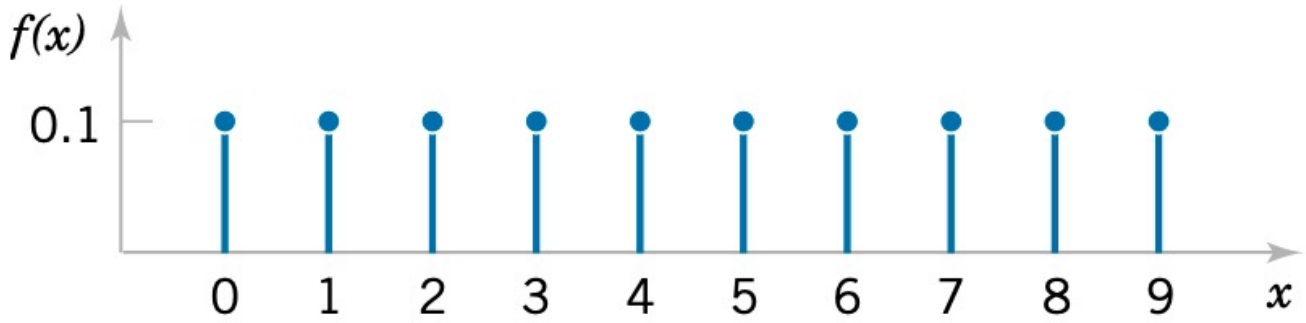
Application: Consider a collection of $N = N_1 + N_2$ similar objects, N_1 of them belonging to one of two dichotomous classes (red chips, say) and *N*2 of them belonging to the second class (blue chips, say). A collection of *n* objects is selected from these *N* objects at random and without replacement. Find the probability that exactly *x* of these *n* objects belong to the first class and *n* − *x* belong to the second.

pmf, mean and variance:

$$f(x) = \frac{\binom{N_1}{x}\binom{N_2}{n-x}}{\binom{N_1+N_2}{n}} \qquad N = N_1 + N_2$$

$$\mu = E(X) = n\left(\frac{N_1}{N}\right) \qquad \sigma^2 = n\left(\frac{N_1}{N}\right)\left(\frac{N_2}{N}\right)\left(\frac{N-n}{N-1}\right)$$

Uniform Distribution



pmf, mean and variance:

$$f(x) = \frac{1}{n}, \qquad x = x_1, x_2, \ldots, x_n$$

$$\mu = E(X) = \frac{1}{n}\sum x_i \qquad \sigma^2 = E(X^2) - [E(X)]^2 = \frac{1}{n}\sum x_i^2 - \left(\frac{1}{n}\sum x_i\right)^2$$

Poisson Distribution

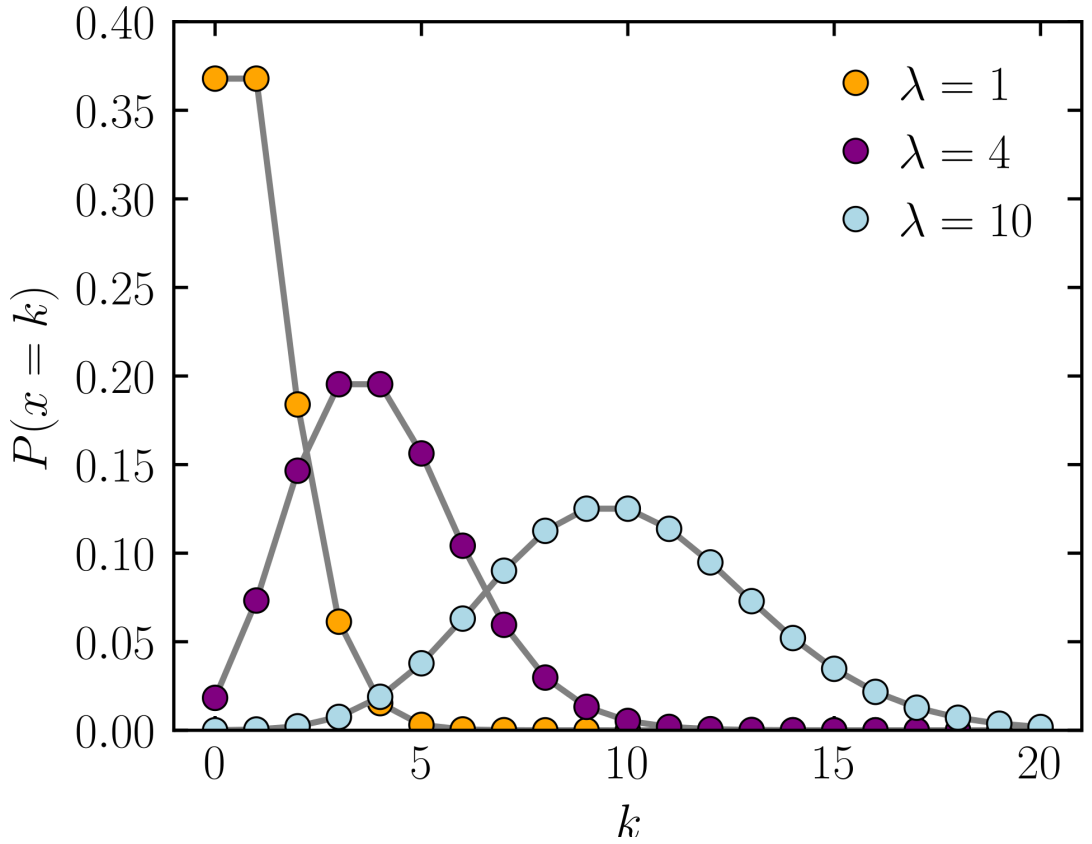
$$X \sim Poi(\lambda) \quad x = 0,1,2, \ldots$$

pmf, mean and variance:

$$f(x) = \frac{\lambda^x e^{-\lambda}}{x!}, \qquad x = 0, 1, 2, \ldots$$

$$E(X) = \sum_{x=0}^{\infty} x \frac{\lambda^x e^{-\lambda}}{x!} = \lambda$$

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = \lambda$$



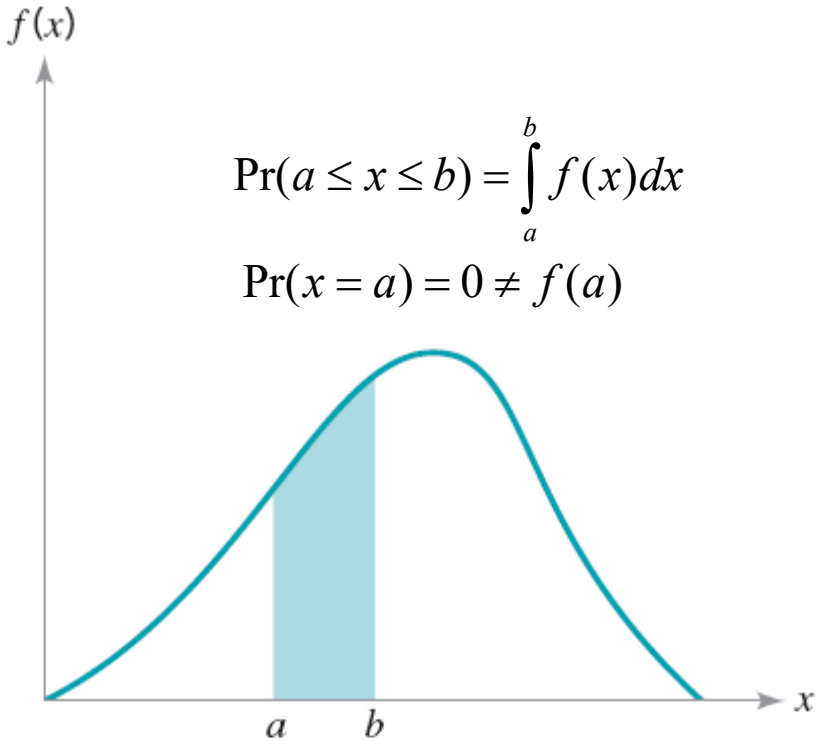
Probability Distribution and Parameter Values	Probability Mass Function	Moment-Generating Function	Mean $E(X)$	Variance $\mathrm{Var}(X)$	Examples
Bernoulli $0 < p < 1$ $q = 1 - p$	$p^x q^{1-x}, \quad x = 0, 1$	$q + pe^t,$ $-\infty < t < \infty$	p	pq	Experiment with two possible outcomes, say success and failure, $p = P(\text{success})$
Binomial $n = 1, 2, 3, \ldots$ $0 < p < 1$	$\binom{n}{x} p^x q^{n-x},$ $x = 0, 1, \ldots, n$	$(q + pe^t)^n,$ $-\infty < t < \infty$	np	npq	Number of successes in a sequence of n Bernoulli trials, $p = P(\text{success})$
Geometric $0 < p < 1$ $q = 1 - p$	$q^{x-1} p,$ $x = 1, 2, \ldots$	$\dfrac{pe^t}{1 - qe^t}$ $t < -\ln(1 - p)$	$\dfrac{1}{p}$	$\dfrac{q}{p^2}$	The number of trials to obtain the first success in a sequence of Bernoulli trials
Hypergeometric $x \le n, x \le N_1$ $n - x \le N_2$ $N = N_1 + N_2$ $N_1 > 0, \quad N_2 > 0$	$\dfrac{\binom{N_1}{x}\binom{N_2}{n-x}}{\binom{N}{n}}$		$n\left(\dfrac{N_1}{N}\right)$	$n\left(\dfrac{N_1}{N}\right)\left(\dfrac{N_2}{N}\right)\left(\dfrac{N-n}{N-1}\right)$	Selecting n objects at random without replacement from a set composed of two types of objects
Negative Binomial $r = 1, 2, 3, \ldots$ $0 < p < 1$	$\binom{x-1}{r-1} p^r q^{x-r},$ $x = r, r+1, \ldots$	$\dfrac{(pe^t)^r}{(1 - qe^t)^r},$ $t < -\ln(1 - p)$	$\dfrac{r}{p}$	$\dfrac{rq}{p^2}$	The number of trials to obtain the rth success in a sequence of Bernoulli trials
Poisson $\lambda > 0$	$\dfrac{\lambda^x e^{-\lambda}}{x!},$ $x = 0, 1, \ldots$	$e^{\lambda(e^t - 1)}$ $-\infty < t < \infty$	λ	λ	Number of events occurring in a unit interval, events are occurring randomly at a mean rate of λ per unit interval
Uniform $m > 0$	$\dfrac{1}{m}, \quad x = 1, 2, \ldots, m$		$\dfrac{m+1}{2}$	$\dfrac{m^2 - 1}{12}$	Select an integer randomly from $1, 2, \ldots, m$

Continuous Distributions

Continuous Distributions

- When the outcome takes continuous values

Probability density function (pdf): $f(x)$ is pdf $\leftrightarrow \begin{cases} f(x) \geq 0 \\ \int_{-\infty}^{+\infty} f(x)dx = 1 \end{cases}$



$$\Pr(a \leq x \leq b) = \int_a^b f(x)dx$$

$$\Pr(x = a) = 0 \neq f(a)$$

CDF: $F(x) = \int_{-\infty}^{x} f(u)du$

Mean: $E(X) = \int xf(x)dx = \mu$

Variance: $V(X) = E(X^2) - [E(X)]^2$

Moment Generating Function (MGF)

- The moment generating function of $X \sim f(x)$, if exists, is defined as

$$M(t) = \int_{-\infty}^{\infty} e^{tx} f(x)\, dx, \qquad -h < t < h.$$

$$\mu = E(X) = \int_{-\infty}^{\infty} x f(x)\, dx = M'(0)$$

$$\sigma^2 = \mathrm{Var}(X) = E[(X-\mu)^2] = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\, dx = M''(0) - [M'(0)]^2$$

Example: The probability distribution of X is

$$f(x) = 2e^{-kx}, \ x \geq 0;$$

(a) Find the appropriate value of k;
(b) Find mean of X;
(c) Find the CDF of X;
(d) Find P(a < X < b).

Exponential distribution

$$X \sim Exp\left(\frac{1}{\theta}\right) \qquad f(x) = \frac{1}{\theta} e^{-x/\theta}, \qquad 0 \le x < \infty$$

MGF:
$$M(t) = \int_0^\infty e^{tx} \left(\frac{1}{\theta}\right) e^{-x/\theta}\, dx = \lim_{b\to\infty} \int_0^b \left(\frac{1}{\theta}\right) e^{-(1-\theta t)x/\theta}\, dx$$

$$= \lim_{b\to\infty} \left[-\frac{e^{-(1-\theta t)x/\theta}}{1-\theta t} \right]_0^b = \frac{1}{1-\theta t}, \qquad t < \frac{1}{\theta}.$$

$\Longrightarrow$
$$M'(t) = \frac{\theta}{(1-\theta t)^2} \qquad M''(t) = \frac{2\theta^2}{(1-\theta t)^3}$$

$$\mu = M'(0) = \theta \qquad \sigma^2 = M''(0) - [M'(0)]^2 = \theta^2$$

Exponential distribution

$$f(x) = \frac{1}{\theta} e^{-x/\theta}, \qquad 0 \le x < \infty$$

Lack of Memory Property

For an exponential random variable X,

$$P(X < t_1 + t_2 \mid X > t_1) = P(X < t_2)$$

Note:
For continuous random variables whose support is (0,∞), the exponential distribution is the only continuous-type distribution with this forgetfulness property.

Gamma distribution

$$X \sim Gamma(\alpha, \theta) \qquad f(x) = \frac{1}{\Gamma(\alpha)\theta^{\alpha}} x^{\alpha-1} e^{-x/\theta}, \qquad 0 \le x < \infty$$

Gamma function: $\Gamma(t) = \int_0^{\infty} y^{t-1} e^{-y}\, dy, \qquad 0 < t$

MGF: $$M(t) = \frac{1}{(1-\theta t)^{\alpha}}, \qquad t < 1/\theta$$

$\Longrightarrow$

$$\mu = \alpha\theta$$

$$\sigma^2 = \alpha\theta^2$$

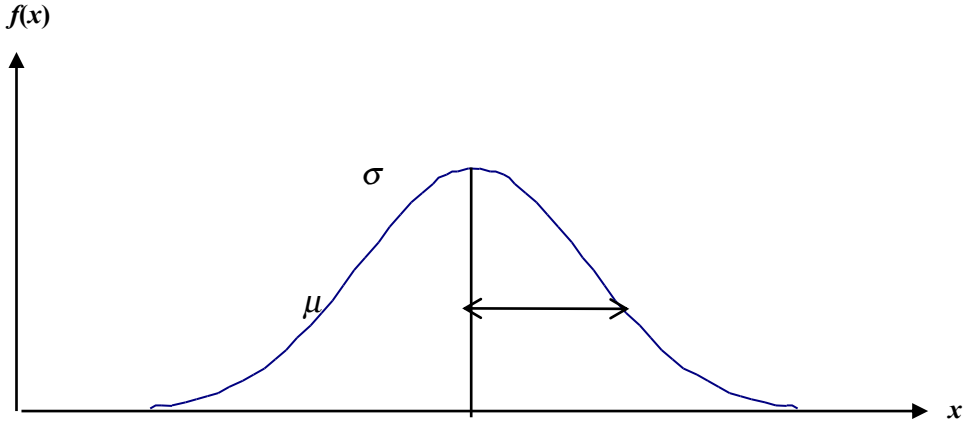
Note:
Exponential distribution is a special case of Gamma distribution with $\alpha = 1$.

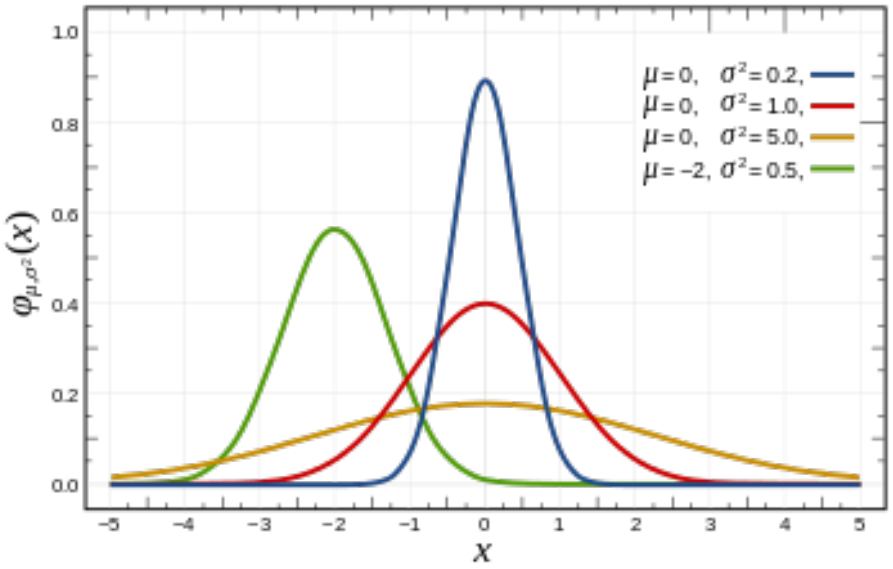
Normal (Gaussian) Distribution

$$X \sim N(\mu,\sigma^2); \qquad -\infty < x < +\infty$$

$$f(x;\mu,\sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$E(x) = \mu \qquad Var(x) = \sigma^2$$



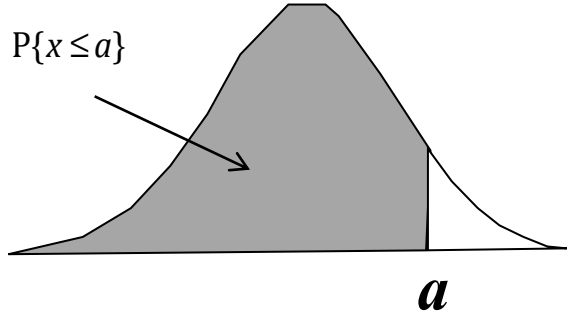




Normal (Gaussian) Distribution

$$f(x;\mu,\sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$\mathrm{P}\{x \le a\} = \int_{-\infty}^{a} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right\} dx$$



MGF: $$M(t) = \exp\left(\frac{2\mu\sigma^2 t + \sigma^4 t^2}{2\sigma^2}\right) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right).$$

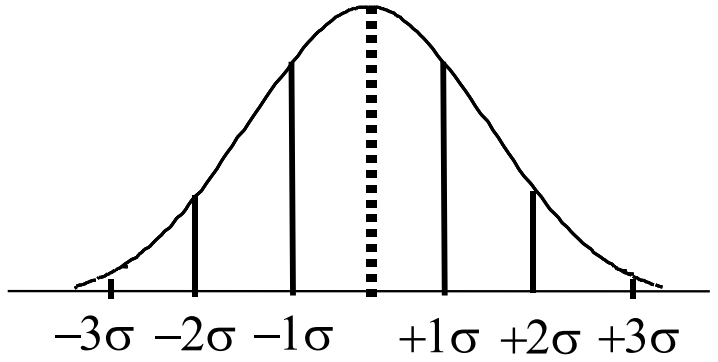
$$E(X) = M'(0) = \mu,$$

$$\mathrm{Var}(X) = M''(0) - [M'(0)]^2 = \mu^2 + \sigma^2 - \mu^2 = \sigma^2$$

Standard Normal Distribution

$$X \sim N(\mu, \sigma^2); \qquad -\infty < x < +\infty$$

$$f(x;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right\}$$

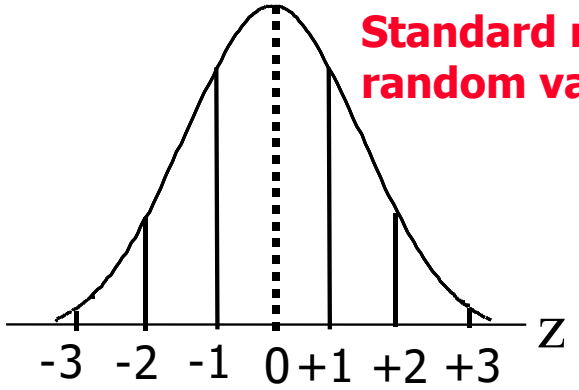


Map any X into Z

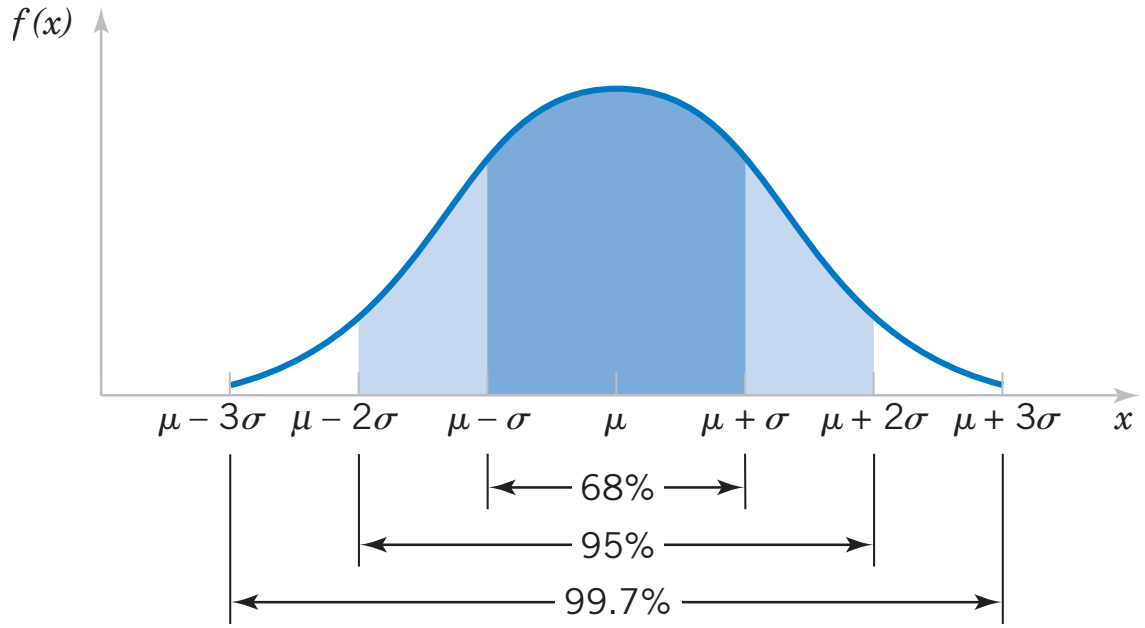
$$Z = \frac{X-\mu}{\sigma}$$

$$Z \sim N(0,1^2); \qquad -\infty < z < +\infty$$

$$f(z;\mu=0,\sigma^2=1) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right)$$



Standard normal random variable



$$\begin{aligned}
\mathbb{P}(\mu - \sigma \leq X \leq \mu + \sigma) &= \mathbb{P}(-1 \leq Z \leq 1) = 0.6827, \\
\mathbb{P}(\mu - 2\sigma \leq X \leq \mu + 2\sigma) &= \mathbb{P}(-2 \leq Z \leq 2) = 0.9545, \\
\mathbb{P}(\mu - 3\sigma \leq X \leq \mu + 3\sigma) &= \mathbb{P}(-3 \leq Z \leq 3) = 0.9973.
\end{aligned}$$

Standard Normal Distribution

$$Z \sim N(0,1)$$

CDF: $\Phi(z) = P(Z \leq z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-w^2/2}\, dw$

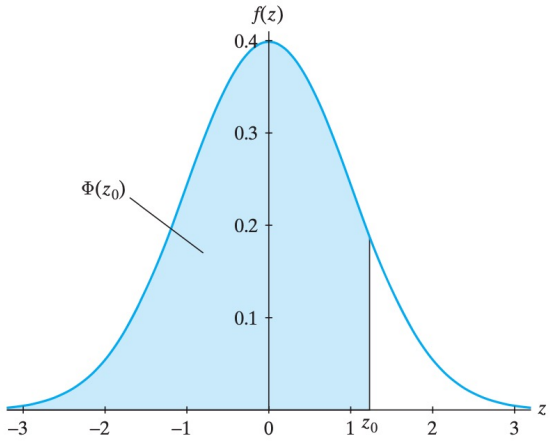


Figure 3.3-1 Standard normal pdf

Z-Values

Quantile: interested in finding a number z_α:

$$P(Z \geq z_\alpha) = \alpha$$

$$P(Z \leq -z_\alpha) = P(Z \geq z_\alpha) = \alpha$$

$$\Longrightarrow \quad z_{1-\alpha} = -z_\alpha$$

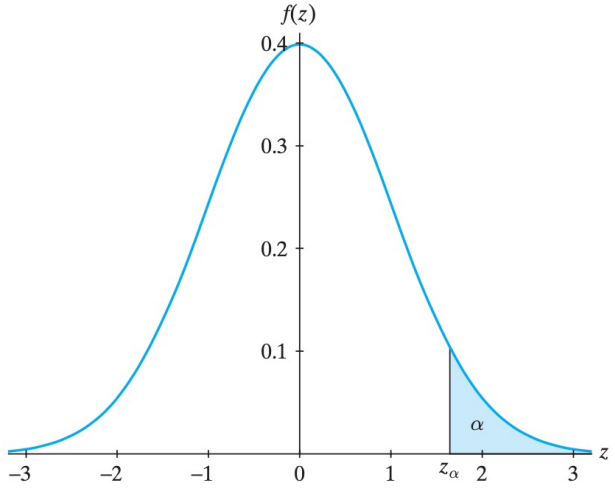
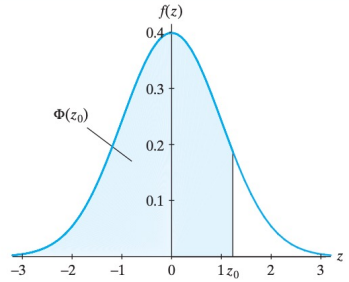


Figure 3.3-2 Upper 100α percent point, z_α

Normal (Gaussian) Distribution

Table Va The Standard Normal Distribution Function



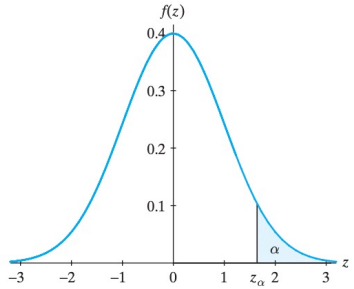
$$P(Z \le z) = \Phi(z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-w^2/2}\, dw$$

$$\Phi(-z) = 1 - \Phi(z)$$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7703	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

α	0.400	0.300	0.200	0.100	0.050	0.025	0.020	0.010	0.005	0.001
z_α	0.253	0.524	0.842	1.282	1.645	1.960	2.054	2.326	2.576	3.090
$z_{\alpha/2}$	0.842	1.036	1.282	1.645	1.960	2.240	2.326	2.576	2.807	3.291

Table Vb The Standard Normal Right-Tail Probabilities



$$P(Z > z_\alpha) = \alpha$$

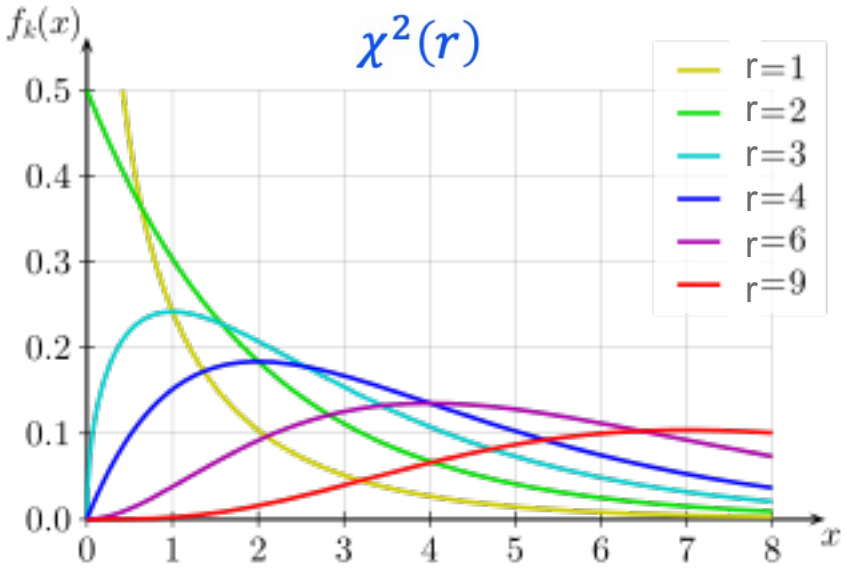
$$P(Z > z) = 1 - \Phi(z) = \Phi(-z)$$

z_α	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002

Chi-square distribution

- Chi-square distribution with r degrees of freedom:

$$X \sim \chi^2(r) \qquad f(x) = \frac{1}{\Gamma(r/2)2^{r/2}} x^{r/2-1} e^{-x/2}, \qquad 0 < x < \infty$$





Chi-square distribution

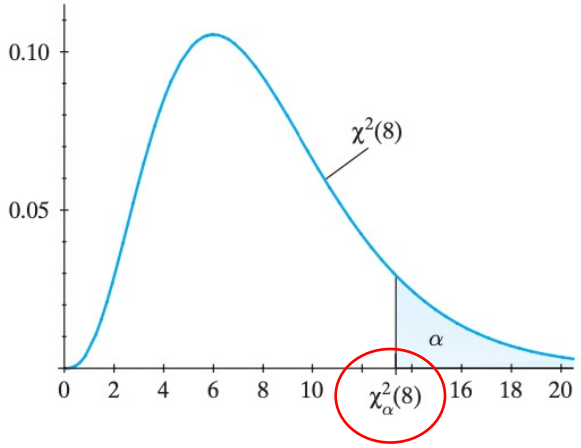
- Chi-square distribution with r degrees of freedom:

$$X \sim \chi^2(r) \qquad f(x) = \frac{1}{\Gamma(r/2)2^{r/2}} x^{r/2-1} e^{-x/2}, \qquad 0 < x < \infty$$

$$Gamma\left(\frac{r}{2}, 2\right)$$

Recall Gamma function: $\Gamma(t) = \int_0^\infty y^{t-1} e^{-y}\, dy, \qquad 0 < t$

$$M(t) = (1-2t)^{-r/2}, \qquad t < \frac{1}{2} \qquad \mu = \alpha\theta = \left(\frac{r}{2}\right)2 = r \qquad \sigma^2 = \alpha\theta^2 = \left(\frac{r}{2}\right)2^2 = 2r$$



	$P(X \le x)$							
	0.010	0.025	0.050	0.100	0.900	0.950	0.975	0.990
r	$\chi^2_{0.99}(r)$	$\chi^2_{0.975}(r)$	$\chi^2_{0.95}(r)$	$\chi^2_{0.90}(r)$	$\chi^2_{0.10}(r)$	$\chi^2_{0.05}(r)$	$\chi^2_{0.025}(r)$	$\chi^2_{0.01}(r)$
1	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.34
4	0.297	0.484	0.711	1.064	7.779	9.488	11.14	13.28
5	0.554	0.831	1.145	1.610	9.236	11.07	12.83	15.09
6	0.872	1.237	1.635	2.204	10.64	12.59	14.45	16.81
7	1.239	1.690	2.167	2.833	12.02	14.07	16.01	18.48
8	1.646	2.180	2.733	3.490	13.36	15.51	17.54	20.09
9	2.088	2.700	3.325	4.168	14.68	16.92	19.02	21.67
10	2.558	3.247	3.940	4.865	15.99	18.31	20.48	23.21
11	3.053	3.816	4.575	5.578	17.28	19.68	21.92	24.72
12	3.571	4.404	5.226	6.304	18.55	21.03	23.34	26.22
13	4.107	5.009	5.892	7.042	19.81	22.36	24.74	27.69
14	4.660	5.629	6.571	7.790	21.06	23.68	26.12	29.14
15	5.229	6.262	7.261	8.547	22.31	25.00	27.49	30.58

Useful Results for Normal and Chi-square

- If $x_1 \sim N(\mu_1, \sigma_1^2), x_2 \sim N(\mu_2, \sigma_2^2)$ are independently normally distributed variables, then

$$y = x_1 + x_2$$

also follows the normal distribution:

$$y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

Theorem If X is $N(\mu, \sigma^2)$, then $Z = (X - \mu)/\sigma$ is $N(0, 1)$.

$$X_1, X_2, \cdots, X_n \sim N(\mu, \sigma^2) \rightarrow \bar{X} \sim N(\mu, \frac{\sigma^2}{n}) \rightarrow \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \sim N(0,1)$$

Sampling distribution of sample mean is normal, when samples are normal

Exercise

- **Example: Three shafts are made and assembled in a machine. The length of each shaft, in centimeters, is distributed as follows:**

  Shaft 1: ~ N (75, 0.09)
  Shaft 2: ~ N (60, 0.16)
  Shaft 3: ~ N (25, 0.25)

  Assume the shafts' length are independent to each other:
  (a) What is the distribution of the linkage?
  (b) What is the probability that the linkage will be longer than 160.5 cm?



Useful Results for Normal and Chi-square

Theorem If the random variable X is $N(\mu, \sigma^2)$, $\sigma^2 > 0$, then the random variable $V = (X-\mu)^2/\sigma^2 = Z^2$ is $\chi^2(1)$.

Facts:

- (connection to normal) $X_1, X_2, \ldots, X_r \sim N(0,1)$, and they are independent, then

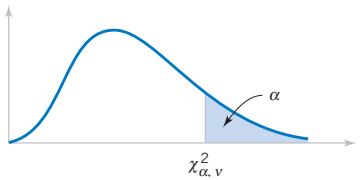
$$X_1^2 + X_2^2 + \cdots + X_r^2 \sim \chi^2(r)$$

- (sum of chi-squares) $X \sim \chi^2(r_1), Y \sim \chi^2(r_2)$, and they are independent, then

$$X + Y \sim \chi^2(r_1 + r_2)$$

Example

- Calculate the following quantities:



1. $\chi^2_{0.05,5}$

2. $\Pr\left\{\sum_{i=1}^{8}\left(\frac{X_i-\mu}{\sigma}\right)^2 > 1.34\right\}$

$X_i \sim N(\mu, \sigma^2)$

3. $\Pr\left\{\left(\frac{X-\mu_x}{\sigma_x}\right)^2 + \left(\frac{Y-\mu_y}{\sigma_y}\right)^2 > 3\right\}$

Table IV Percentage Points $\chi^2_{\alpha,\nu}$ of the Chi-Squared Distribution

ν \ α	.995	.990	.975	.950	.900	.500	.100	.050	.025	.010	.005
1	.00+	.00+	.00+	.00+	.02	.45	2.71	3.84	5.02	6.63	7.88
2	.01	.02	.05	.10	.21	1.39	4.61	5.99	7.38	9.21	10.60
3	.07	.11	.22	.35	.58	2.37	6.25	7.81	9.35	11.34	12.84
4	.21	.30	.48	.71	1.06	3.36	7.78	9.49	11.14	13.28	14.86
5	.41	.55	.83	1.15	1.61	4.35	9.24	11.07	12.83	15.09	16.75
6	.68	.87	1.24	1.64	2.20	5.35	10.65	12.59	14.45	16.81	18.55
7	.99	1.24	1.69	2.17	2.83	6.35	12.02	14.07	16.01	18.48	20.28
8	1.34	1.65	2.18	2.73	3.49	7.34	13.36	15.51	17.53	20.09	21.96
9	1.73	2.09	2.70	3.33	4.17	8.34	14.68	16.92	19.02	21.67	23.59
10	2.16	2.56	3.25	3.94	4.87	9.34	15.99	18.31	20.48	23.21	25.19
11	2.60	3.05	3.82	4.57	5.58	10.34	17.28	19.68	21.92	24.72	26.76
12	3.07	3.57	4.40	5.23	6.30	11.34	18.55	21.03	23.34	26.22	28.30
13	3.57	4.11	5.01	5.89	7.04	12.34	19.81	22.36	24.74	27.69	29.82
14	4.07	4.66	5.63	6.57	7.79	13.34	21.06	23.68	26.12	29.14	31.32

X and Y are independent and follow a normal dist.

$X \sim N(\mu_x, \sigma_x^2), Y \sim N(\mu_y, \sigma_y^2)$

F distribution

- Let $U \sim \chi^2(r_1), V \sim \chi^2(r_2)$, and they are independent

$$W = \frac{U/r_1}{V/r_2} \qquad f(w) = \frac{(r_1/r_2)^{r_1/2}\Gamma[(r_1+r_2)/2]w^{r_1/2-1}}{\Gamma(r_1/2)\Gamma(r_2/2)[1+(r_1 w/r_2)]^{(r_1+r_2)/2}}$$

$W \sim F(r_1, r_2)$: F distribution with r_1 and r_2 degrees of freedom

Facts:

- $W \sim F(r_1, r_2) \xrightarrow{\text{yields}} \frac{1}{W} \sim F(r_2, r_1)$
- Denote right-tail probability $P[W \geq F_\alpha(r_1, r_2)] = \alpha$

$$\alpha = P[W \leq F_{1-\alpha}(r_1, r_2)] = P[1/W \geq 1/F_{1-\alpha}(r_1, r_2)]$$

$$\Longrightarrow \frac{1}{F_{1-\alpha}(r_1, r_2)} = F_\alpha(r_2, r_1) \text{ or } F_{1-\alpha}(r_1, r_2) = \frac{1}{F_\alpha(r_2, r_1)}$$

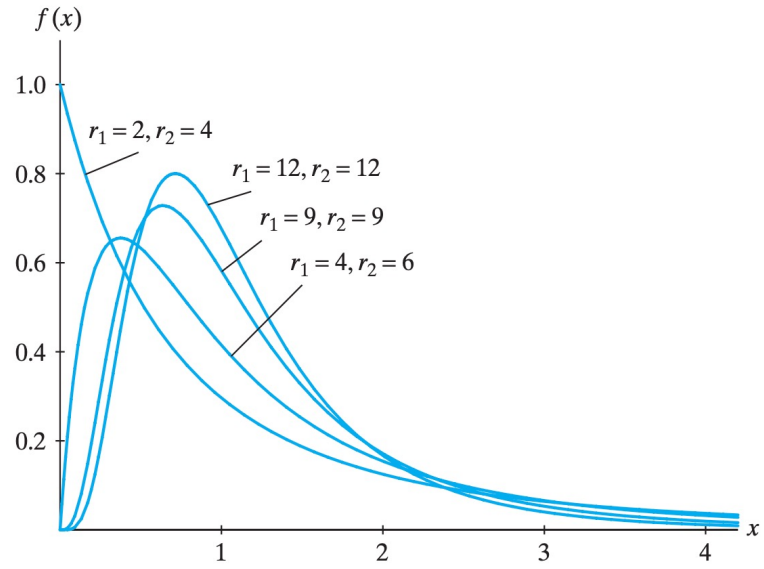


Figure 5.2-5 Graphs of F pdfs

Exercise

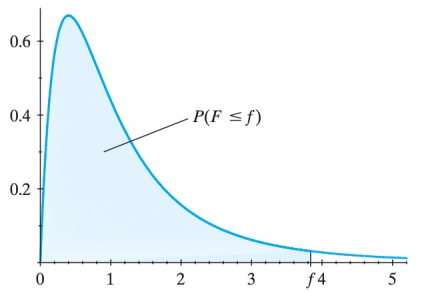
- Calculate the following quantities:

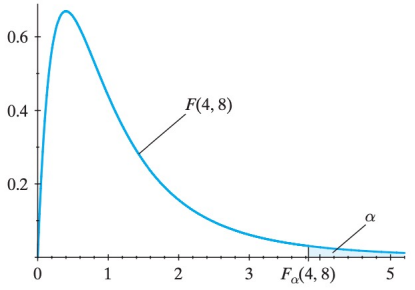
1. $F_{0.025}(6,8)$

2. $X \sim \mathcal{N}(\mu_x, \sigma_x^2)$

$$Y_1, Y_2 \sim \mathcal{N}(\mu_y, \sigma_y^2)$$

$$\mathbb{P}\left\{ \frac{2\left(\frac{X-\mu_x}{\sigma_x}\right)^2}{\left(\frac{Y_1-\mu_y}{\sigma_y}\right)^2 + \left(\frac{Y_2-\mu_y}{\sigma_y}\right)^2} > w \right\}$$
$$= \mathbb{P}\{F(1,2) > w\}$$





$$P(F \le f) = \int_0^f \frac{\Gamma[(r_1+r_2)/2](r_1/r_2)^{r_1/2} w^{r_1/2-1}}{\Gamma(r_1/2)\Gamma(r_2/2)(1+r_1 w/r_2)^{(r_1+r_2)/2}}\, dw$$

α	$P(F \le f)$	Den. d.f. r_2	Numerator Degrees of Freedom, r_1									
			1	2	3	4	5	6	7	8	9	10
0.05	0.95	1	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5	241.9
0.025	0.975		647.79	799.50	864.16	899.58	921.85	937.11	948.22	956.66	963.28	968.63
0.01	0.99		4052	4999.5	5403	5625	5764	5859	5928	5981	6022	6056
0.05	0.95	2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40
0.025	0.975		38.51	39.00	39.17	39.25	39.30	39.33	39.36	39.37	39.39	39.40
0.01	0.99		98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39	99.40
0.05	0.95	3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79
0.025	0.975		17.44	16.04	15.44	15.10	14.88	14.73	14.62	14.54	14.47	14.42
0.01	0.99		34.12	30.82	29.46	28.71	28.24	27.91	27.67	27.49	27.35	27.23
0.05	0.95	4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96
0.025	0.975		12.22	10.65	9.98	9.60	9.36	9.20	9.07	8.98	8.90	8.84
0.01	0.99		21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80	14.66	14.55
0.05	0.95	5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74
0.025	0.975		10.01	8.43	7.76	7.39	7.15	6.98	6.85	6.76	6.68	6.62
0.01	0.99		16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29	10.16	10.05
0.05	0.95	6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06
0.025	0.975		8.81	7.26	6.60	6.23	5.99	5.82	5.70	5.60	5.52	5.46
0.01	0.99		13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10	7.98	7.87
0.05	0.95	7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64
0.025	0.975		8.07	6.54	5.89	5.52	5.29	5.12	4.99	4.90	4.82	4.76
0.01	0.99		12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62
0.05	0.95	8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35
0.025	0.975		7.57	6.06	5.42	5.05	4.82	4.65	4.53	4.43	4.36	4.30
0.01	0.99		11.26	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81
0.05	0.95	9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14
0.025	0.975		7.21	5.71	5.08	4.72	4.48	4.32	4.20	4.10	4.03	3.96
0.01	0.99		10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26
0.05	0.95	10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98
0.025	0.975		6.94	5.46	4.83	4.47	4.24	4.07	3.95	3.85	3.78	3.72
0.01	0.99		10.04	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85

Student's t distribution

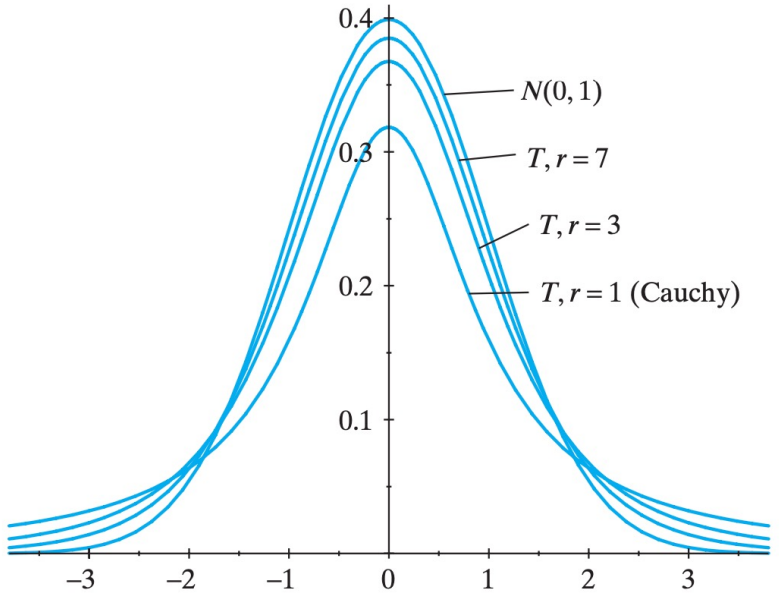
- Let $Z \sim N(0,1), U \sim \chi^2(r)$, and they are independent

$$T = \frac{Z}{\sqrt{U/r}} \qquad f(t) = \frac{\Gamma((r+1)/2)}{\sqrt{\pi r}\,\Gamma(r/2)} \frac{1}{(1+t^2/r)^{(r+1)/2}}$$

$T \sim t(r)$: t distribution with r degrees of freedom

Facts:

The larger the degrees of freedom r, the more the t distribution looks like the normal distribution



Example

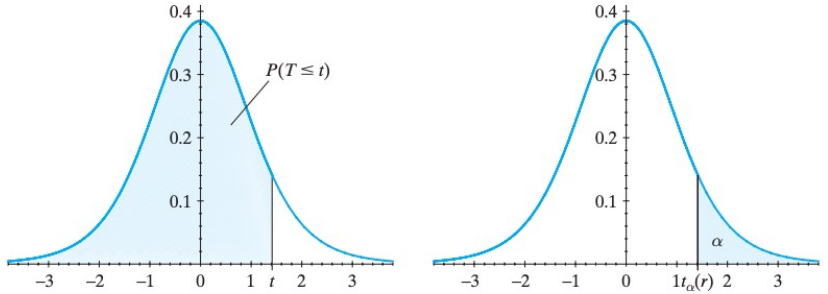
- Calculate the following quantities:

1. $t_{0.05,5}$

2. $\Pr\left(\frac{Z}{\sqrt{\left(\frac{X-\mu_x}{\sigma_x}\right)^2}} > 1\right)$

Z follows a standard normal dist. and X follows a normal dist. and they are independent

Table VI The t Distribution



$$P(T \le t) = \int_{-\infty}^{t} \frac{\Gamma[(r+1)/2]}{\sqrt{\pi r}\,\Gamma(r/2)(1+w^2/r)^{(r+1)/2}}\,dw$$

$$P(T \le -t) = 1 - P(T \le t)$$

	$P(T \le t)$						
	0.60	0.75	0.90	0.95	0.975	0.99	0.995
r	$t_{0.40}(r)$	$t_{0.25}(r)$	$t_{0.10}(r)$	$t_{0.05}(r)$	$t_{0.025}(r)$	$t_{0.01}(r)$	$t_{0.005}(r)$
1	0.325	1.000	3.078	6.314	12.706	31.821	63.657
2	0.289	0.816	1.886	2.920	4.303	6.965	9.925
3	0.277	0.765	1.638	2.353	3.182	4.541	5.841
4	0.271	0.741	1.533	2.132	2.776	3.747	4.604
5	0.267	0.727	1.476	2.015	2.571	3.365	4.032
6	0.265	0.718	1.440	1.943	2.447	3.143	3.707
7	0.263	0.711	1.415	1.895	2.365	2.998	3.499
8	0.262	0.706	1.397	1.860	2.306	2.896	3.355
9	0.261	0.703	1.383	1.833	2.262	2.821	3.250
10	0.260	0.700	1.372	1.812	2.228	2.764	3.169

Probability Distribution and Parameter Values	Probability Density Function	Moment-Generating Function	Mean $E(X)$	Variance $\mathrm{Var}(X)$	Examples
Beta $\alpha > 0$ $\beta > 0$	$\dfrac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}$, $0 < x < 1$		$\dfrac{\alpha}{\alpha+\beta}$	$\dfrac{\alpha\beta}{(\alpha+\beta+1)(\alpha+\beta)^2}$	$X = X_1/(X_1+X_2)$, where X_1 and X_2 have independent gamma distributions with same θ
Chi-square $r = 1, 2, \ldots$	$\dfrac{x^{r/2-1}e^{-x/2}}{\Gamma(r/2)2^{r/2}}$, $0 < x < \infty$	$\dfrac{1}{(1-2t)^{r/2}}$, $t < \dfrac{1}{2}$	r	$2r$	Gamma distribution, $\theta = 2$, $\alpha = r/2$; sum of squares of r independent $N(0,1)$ random variables
Exponential $\theta > 0$	$\dfrac{1}{\theta} e^{-x/\theta}$, $0 \le x < \infty$	$\dfrac{1}{1-\theta t}$, $t < \dfrac{1}{\theta}$	θ	θ^2	Waiting time to first arrival when observing a Poisson process with a mean rate of arrivals equal to $\lambda = 1/\theta$
Gamma $\alpha > 0$ $\theta > 0$	$\dfrac{x^{\alpha-1}e^{-x/\theta}}{\Gamma(\alpha)\theta^{\alpha}}$, $0 < x < \infty$	$\dfrac{1}{(1-\theta t)^{\alpha}}$, $t < \dfrac{1}{\theta}$	$\alpha\theta$	$\alpha\theta^2$	Waiting time to αth arrival when observing a Poisson process with a mean rate of arrivals equal to $\lambda = 1/\theta$
Normal $-\infty < \mu < \infty$ $\sigma > 0$	$\dfrac{e^{-(x-\mu)^2/2\sigma^2}}{\sigma\sqrt{2\pi}}$, $-\infty < x < \infty$	$e^{\mu t+\sigma^2 t^2/2}$ $-\infty < t < \infty$	μ	σ^2	Errors in measurements; heights of children; breaking strengths
Uniform $-\infty < a < b < \infty$	$\dfrac{1}{b-a}$, $a \le x \le b$	$\dfrac{e^{tb}-e^{ta}}{t(b-a)}$, $t \ne 0$ 1, $t = 0$	$\dfrac{a+b}{2}$	$\dfrac{(b-a)^2}{12}$	Select a point at random from the interval $[a, b]$

Find Percentage Points using R

- get percentage points either from statistical tables or using R

- In R, the commands qnorm, qchisq, qt, and qf will get percentage points of the normal, χ^2, *t*, and *F* distributions respectively.

normal	qnorm
χ^2	qchisq
t	qt
F	qf

- qnorm(0.05, lower.tail = F)
- [1] 1.644854
- > qt(0.05, df=3, lower.tail=F)
- [1] 2.353363
- > qchisq(0.05,5,lower.tail=F)
- [1] 11.0705
- > qf(0.25,6,8,lower.tail=F)
- [1] 1.650839

Several random variables

Useful Results on Mean and Variance

If x is a random variable and a is a constant, then

$E(a+x)=a+E(x)$

$E(a \times x)=aE(x)$

$V(a+x)=V(x)$

$V(a \times x)=a^2V(x)$

If $x_1, x_2, \ldots, x_n$ are random variables,

$E(a_1x_1+\ldots+a_nx_n)=a_1E(x_1)+\ldots+a_nE(x_n)$

If they are **mutually independent**, and $a_1,\ldots,a_n$ are constants

$V(a_1x_1+\ldots+a_nx_n)=a_1^2V(x_1)+\ldots+a_n^2V(x_n)$

Covariance and correlation

- Let X, Y be two random variables with mean μ_X, μ_Y, and variance σ_X^2, σ_Y^2
- The **covariance** between X and Y is

$$\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = \sigma_{XY}$$

- The **correlation coefficient** between X and Y is

$$\rho = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

- Fact: if X and Y are independent, then $\sigma_{XY} = \rho = 0$

Statistics and Sampling Distribution

Statistics: e.g. sample mean and variance

- Suppose we have a sequence of i.i.d. (independent and identically distributed) random variables $X_1, X_2, \ldots, X_n$.

> A **statistic** is any function of the observations in a random sample.

- e.g., $X_1, X_2, \cdots, X_n \rightarrow \overline{X}, S^2$
  - The **sample mean** is defined as $\overline{X} = \dfrac{X_1 + X_2 + \cdots + X_n}{n}$
  - The **sample variance** is defined $S^2 = \dfrac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$

> The probability distribution of a statistic is called a **sampling distribution.**

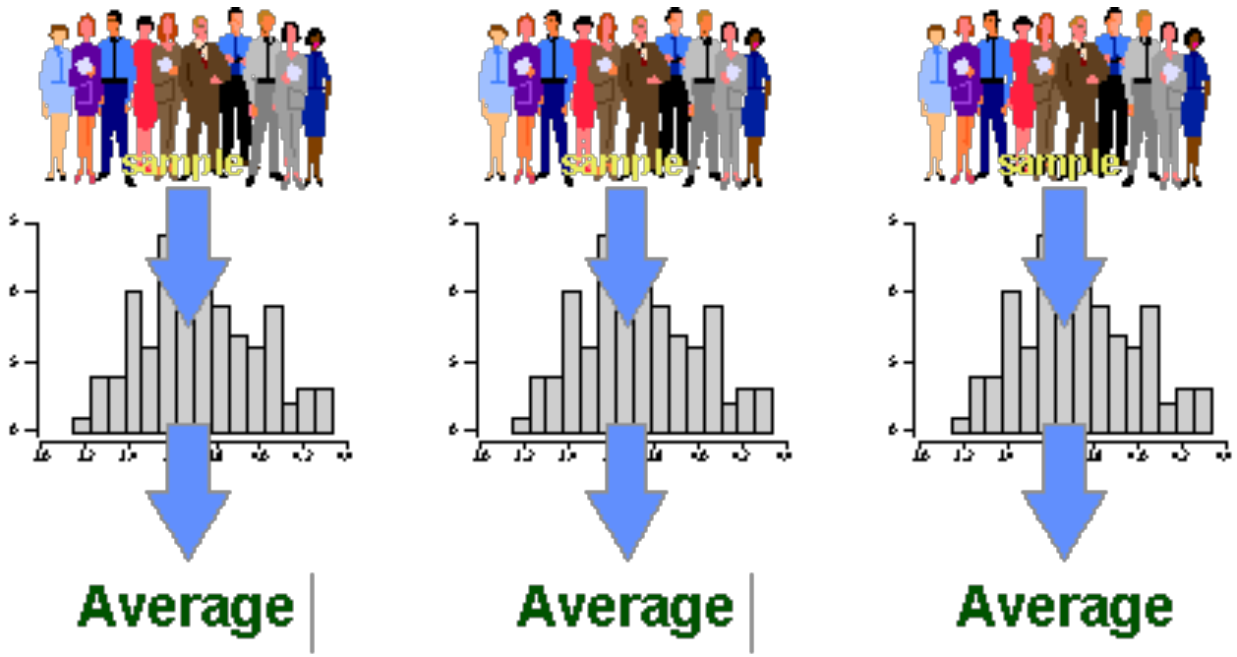
Why sampling distribution?

The probability distribution of a statistic is called a **sampling distribution.**

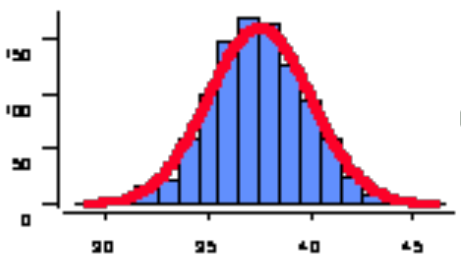
Statistical inference is concerned with making decisions about a population based on the information contained in a *random sample* from that population.

Sampling distribution is the link between probability and statistics.

Sampling distribution



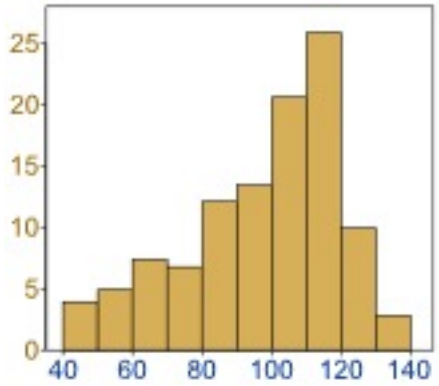
The Sampling Distribution...



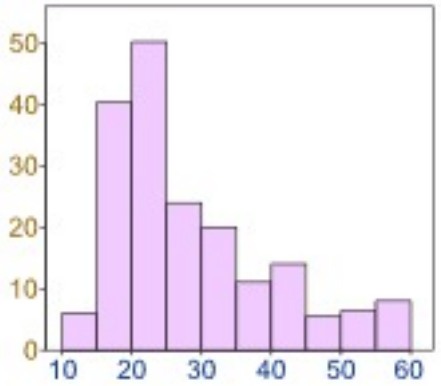
...is the distribution of a statistic across an infinite number of samples

Empirical distribution

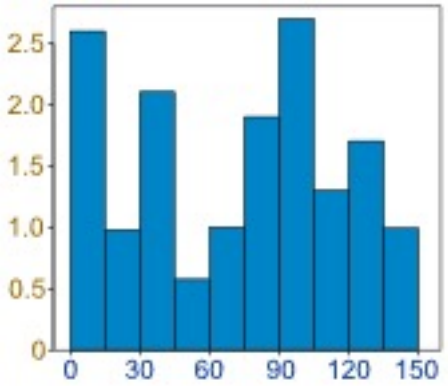
- **Data can be “distributed” (spread out) in different ways**



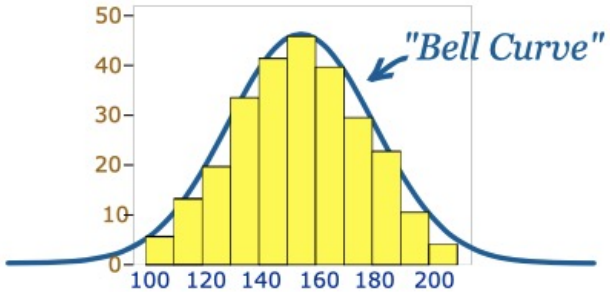
It can be spread out more on the left



Or more on the right



Or it can be all jumbled up



Model sampling distribution

- **Relationships between Bernoulli and Binomial distributions**

$$X_i \sim BERN(p), i = 1, 2, \ldots, n$$

$$\bar{X} = \sum_{i=1}^{n} X_i \sim BIN(n, p)$$





- **In this setting**
- **Each time X_i is the outcome of each draw:**

  = 1, if black, otherwise = 0
- **$\bar{X}$ is the number of black stones**
- **Multiple experiments: $\bar{X}$ is different and has variability**

Alternative view

- **Sample proportion is the percentage of black stones**

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

- **Claim: $\bar{X}$ is approximately normal distributed with mean p and variance = $\frac{p(1-p)}{n}$**

Sampling distribution here describes the distribution of sample mean

Special case

- **Making normal assumption about samples:**

Xi’ s are **normally** independently distributed (a random sample from a Normal distribution with the known variance)

$$X_1, X_2, \cdots, X_n \sim N(\mu, \sigma^2) \rightarrow \bar{X} \sim N(\mu, \frac{\sigma^2}{n}) \rightarrow \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \sim N(0,1)$$

Sampling distribution of sample mean is normal, when samples are normal

Sample mean and variance

Corollary 5.5-1 If $X_1, X_2, \ldots, X_n$ are observations of a random sample of size n from the normal distribution $N(\mu, \sigma^2)$, then the distribution of the sample mean $\overline{X} = (1/n)\sum_{i=1}^{n} X_i$ is $N(\mu, \sigma^2/n)$.

Theorem 5.5-2 Let $X_1, X_2, \ldots, X_n$ be observations of a random sample of size n from the normal distribution $N(\mu, \sigma^2)$. Then the sample mean,

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i,$$

and the sample variance,

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n} (X_i - \overline{X})^2,$$

are independent and

$$\frac{(n-1)S^2}{\sigma^2} = \frac{\sum_{i=1}^{n} (X_i - \overline{X})^2}{\sigma^2} \text{ is } \chi^2(n-1).$$

Example

1. **The design of the machine has fill volume 300 mls, and variance 9ml. An engineer takes a random sample of 25 cans, what's the sampling distribution of mean filling volume of a can of soft drink?**

2. **The engineer finds the sample mean of fill volume to be 298 mls. Is this considered to be normal?**



Exercise: Airport Check-in

The amount of time that a customer spends waiting in the airport check-in counter is a normal random variable with mean 8.2 minutes and standard deviation 1.5 minutes. Suppose that a random sample of 49 customers is observed.



1. Find the probability that the average waiting time for these customers is:
(a) Less than 10 minutes;
(b) Between 5 and 10 minutes.

2. What is a value such that 90% of chance, average wait time will wait shorter than that?

Example

Patient's Waiting Time:

Suppose the patient's waiting time in an ER is normally distributed with σ = 0.5 min. If the waiting time of 3 randomly selected patients are recorded, calculate the probability that the sample standard deviation is greater than 0.5.

Sample mean and variance

- As a consequence, we have

$$U = \sum_{i=1}^{n} \frac{(X_i - \mu)^2}{\sigma^2} \sim \chi^2(n) \qquad W = \sum_{i=1}^{n} \frac{(X_i - \overline{X})^2}{\sigma^2} \sim \chi^2(n-1)$$

$$T = \frac{\overset{N(0,1)}{\overline{X} - \mu}}{\sqrt{\underset{\chi^2(n-1)}{\frac{(n-1)S^2}{\sigma^2}} \Big/ (n-1)}} = \frac{\overline{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$$

Summary: distribution of sample mean

- With known variance (σ^2)

$$\frac{\overline{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

- With unknown variance (σ^2)

$$T = \frac{\frac{\overline{X}-\mu}{\sigma/\sqrt{n}}}{\sqrt{\frac{(n-1)S^2}{\sigma^2} \Big/ (n-1)}} = \frac{\overline{X}-\mu}{S/\sqrt{n}} \sim t(n-1)$$

Note: This result will be used in Chapter 7.

Distribution of sample mean differences

Two Populations: Two independent random samples from two Normal distributions

$$X_1, \ldots, X_{n_1} \sim N(\mu_1, \sigma_1^2) \qquad Y_1, \ldots, Y_{n_2} \sim N(\mu_2, \sigma_2^2)$$

- With known variances:

$$\frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0,1)$$

- With unknown (but equal) variances

$$\frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t(n_1 + n_2 - 2)$$

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{(n_1 + n_2 - 2)}$$

Pooled estimator of variance

Note: This result will be used in Chapter 7.

Example

$n_1 = 10 \quad \bar{x} = 28 \quad S_1^2 = 4$

$n_2 = 10 \quad \bar{y} = 26 \quad S_2^2 = 5$

Assume $\mu_x - \mu_y = -1$

Compute $P(\bar{x}\text{-}\bar{y}>0)$

$$t = \frac{\bar{x}\text{-}\bar{y}}{S_p\sqrt{1/n_1 + 1/n_2}}$$

Compare two drugs



$$S_p^2 = \frac{S_1^2(n_1-1)+S_2^2(n_1-1)}{n_1+n_2-2} = \frac{4(9)+5(9)}{18} = 4.5$$

Sampling Distribution of Sample Variance

Two Populations:

Two independent random samples from two Normal distributions

$$\left.\begin{array}{l} X_1, X_2, \cdots, X_{n_1} \sim N(\mu_1, \sigma_1^2) \\ \\ Y_1, Y_2, \cdots, Y_{n_2} \sim N(\mu_2, \sigma_2^2) \end{array}\right\} \rightarrow \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2} \sim F(n_1 - 1, n_2 - 1)$$

Compare the sample variances of two population

Note: This result will be used in Chapter 7.

Exercise

- Calculate the following quantities:

$$\left.\begin{array}{l} X_1, X_2, X_3 \sim N(\mu_1, \sigma_1^2 = 4) \\ \\ Y_1, Y_2, Y_3 \sim N(\mu_2, \sigma_2^2 = 16) \end{array}\right\} \rightarrow \Pr\left(\frac{S_1^2}{S_2^2} \geq 2\right) = ?$$

Central Limit Theorem (CLT)

Central Limit Theorem

Sampling distribution of sample mean is approximately normal, even when samples are NOT normal

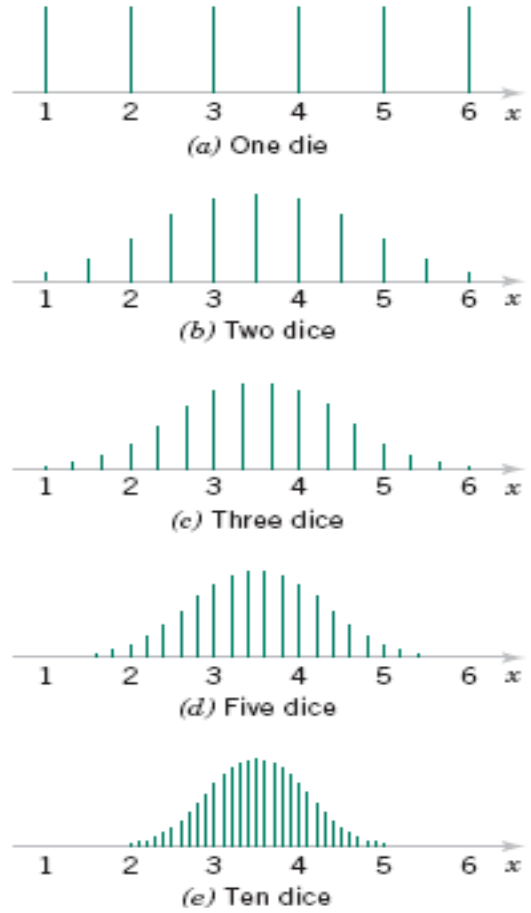


Figure: Distributions of average scores from throwing dice. [Adapted with permission from Box, Hunter, and Hunter (1978).]

Rule of thumb: when n > 30 this works pretty well.

(Informal)

CENTRAL LIMIT THEOREM If all samples of a particular size are selected from any population, the sampling distribution of the sample mean is approximately a normal distribution. This approximation improves with larger samples.

Central Limit Theorem

Theorem 5.6-1 **(Central Limit Theorem)** If $\overline{X}$ is the mean of a random sample $X_1, X_2, \ldots, X_n$ of size n from a distribution with a finite mean μ and a finite positive variance σ^2, then the distribution of

$$W = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \frac{\sum_{i=1}^{n} X_i - n\mu}{\sqrt{n}\,\sigma}$$

is $N(0,1)$ in the limit as $n \to \infty$.

[**Law of large numbers**, Chapter 5.8 in book] Suppose that $X_1, X_2, \ldots, X_n$ are i.i.d. with common mean μ and variance $\sigma^2 < \infty$. Then for any $\epsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}(|\overline{X} - \mu| > \epsilon) = 0.$$

Exercise: Coffee

For the “number of coffee drink / day” question, assume the number of coffee drink / day is a random variable with mean 1 and variance 0.5.

There are 58 responses of survey. What the sampling distribution of the sample mean?



Exercise: Coffee revisit

For the "number of coffee drink / day" question, n = 58. Assume the number of coffee drink / day is a random variable with mean b and unknown variance.

1. **From our study, the sample mean is 0.6636, and the sample std is 0.8638.**
2. **What is the chance for us to observe sample mean great than 1, if the true mean is equal to 0.6636?**



Example: proportion

- Sample proportion is the percentage of success

$$\hat{p} = \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

$X \sim Bernoulli(p)$ $x = 0,1$ p: probability of success

Application: To model the outcome of experiment with two possible outcomes (Success = 1, Failure = 0).

e.g. Flip a coin:  **or**  **?**

- Claim: $\bar{X}(\hat{p})$ is approximately normal distributed with mean p and variance = $\frac{p(1-p)}{n}$

Central Limit Theorem (CLT) for two populations

If we have two independent populations with means μ_1 and μ_2 and variances σ_2^2 and σ_2^2 and if $\bar{X}_1$ and $\bar{X}_2$ are the sample means of two independent random samples of sizes n_1 and n_2 from these populations, then the sampling distribution of

$$Z = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$$

is approximately standard normal, if the conditions of the central limit theorem apply. If the two populations are normal, the sampling distribution of Z is exactly standard normal.

Sampling Distributions

Sampling Distribution: The probability distribution of a statistics based on a random sample

Sampling distributions for sample mean and variance:

- Normal
- Chi-square distribution
- t-distribution
- F-distribution
- Sample proportion
- Comparing two samples

Key issues:

- When should each distribution be used?
- How to relate this to answer questions (statistical inference)?

Summary of Sampling Distributions

		One population	Two populations
Sample Mean	Known Variance	$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$	$\frac{(\bar{X}_1-\bar{X}_2)-(\mu_1-\mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}} \sim N(0,1)$
	Unknown Variance	$\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t(n-1)$	$\frac{(\bar{X}_1-\bar{X}_2)-(\mu_1-\mu_2)}{S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \sim t(n_1+n_2-2)$
Sample Variance		$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$	$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1-1,n_2-1)$
Sample proportion		$\hat{p} \sim N(p, p(1-p)/n)$	$\frac{(\hat{p}_1-\hat{p}_2)-(p_1-p_2)}{\sqrt{\frac{p_1(1-p_1)}{n_1}+\frac{p_2(1-p_2)}{n_2}}} \sim N(0,1)$

Summary

Q: Why do we need these distributions?

A: They are introduced to characterize randomness in our samples.

More to come.

